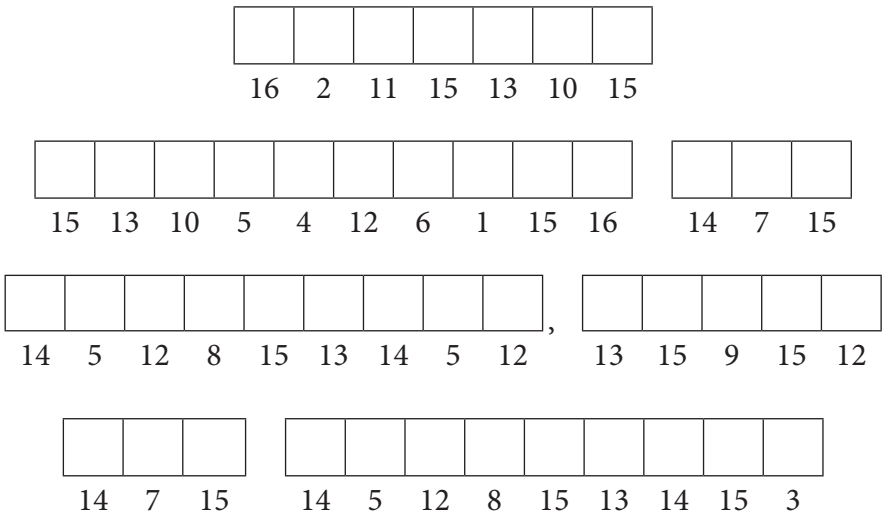
Name: _____

WORKSHEET

The cosine rule

I swore never to be silent whenever and wherever human beings endure suffering and humiliation. We must always take sides. Neutrality helps the oppressor, never the victim …



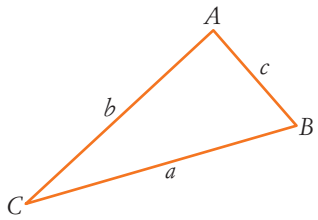
Elie Wiesel

Rule

The cosine rule can be used to solve for unknown sides or angles of non-right-angled triangles when we know:

- all three side lengths, or
- two side lengths and the included angle.

$a^2 = b^2 + c^2 - 2bc \cos A$
$b^2 = c^2 + a^2 - 2ca \cos B$
$c^2 = a^2 + b^2 - 2ab \cos C$

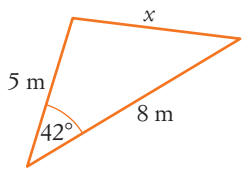


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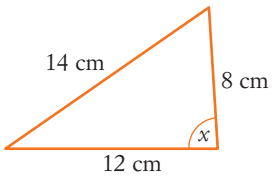
Questions

Find the unknown side or angle indicated in each triangle.

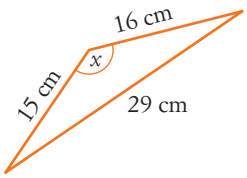
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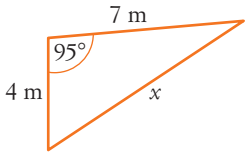
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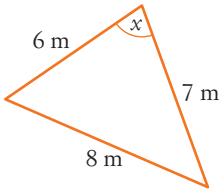
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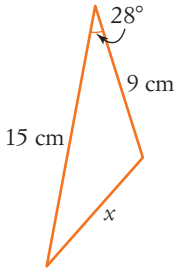
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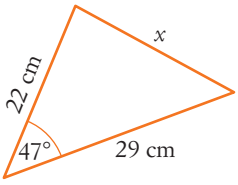
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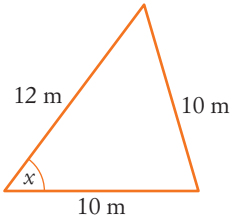
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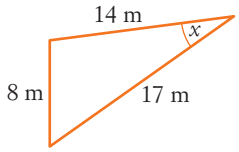
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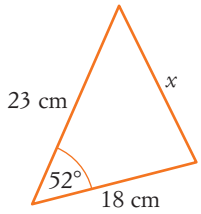
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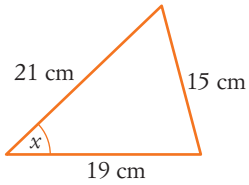
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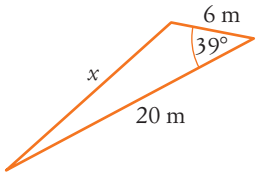
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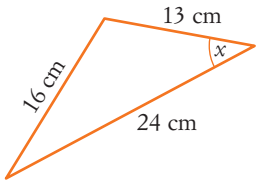
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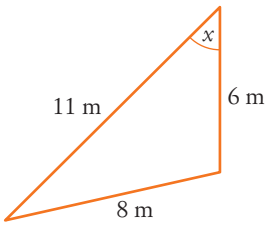
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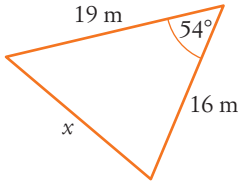
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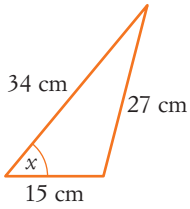
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15



16



Solutions

A 8.2 cm	**G** 5.4 m	**M** 53.1°	**S** 50.3°
C 18.5 cm	**H** 21.3 cm	**N** 38.4°	**T** 45.2°
D 138.6°	**I** 86.4°	**O** 75.5°	**U** 8.4 m
E 16.1 m	**L** 43.7°	**R** 15.8 m	**V** 27.8°

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